

CSE370: Database Systems
Spring 2026
Total Marks: 20 Marks
Duration - 20 minutes

Name:

Section:

Student ID:

Question 1 [10 Marks]

Playlist (listenerID, songID, albumID, listenerName, subscriptionType, region, songTitle, genre, duration, albumTitle, releaseYear, artistName, artistLabel, artistEmail, playTime, rating)

The primary key is underlined.

The relation has the following additional functional dependencies:

FD1: listenerID → listenerName, subscriptionType, region

FD2: songID → songTitle, genre, duration

FD3: albumID → albumTitle, releaseYear

FD4: songID, albumID → artistName, artistLabel, artistEmail

FD5: artistName → artistLabel, artistEmail

- a. Explain if this is in **1NF** or not. If not, decompose it to 1NF. [2]
- b. Explain if this is in **2NF** or not. If not, decompose it to 2NF. [4]
- c. Explain if this is in **3NF** or not. If not, decompose it to 3NF. [4]

- a. Already in 1NF as there are no composite, multivalued attributes, and no nested relations present.
- b. NOT in 2NF due to partial dependency of FD1, FD2, FD3, FD4
Decomposing to 2NF:
Listeners (listenerID, listenerName, subscriptionType, region)
Songs (songID, songTitle, genre, duration)
Albums (albumID, albumTitle, releaseYear)
AlbumSong (songID, albumID, artistName, artistLabel, artistEmail)
Playlist (listenerID, songID, albumID, playTime, rating)
- c. NOT in 3NF due to TD of FD5
AlbumSong (songID, albumID, artistName)
Artist (artistName, artistLabel, artistEmail)

The rest remains unchanged

Question 2 [10 Marks]

Computer_Repair(Comp_ID,Engineer_ID,Date_Assigned,Customer_name, Customer_phone, Engineer_Name, Engineer_phone, Date_Repaired, Issue, Priority_Level, Service_Charge, Commission_Percentage, Total_Repairs)

The primary key of the relation is underlined

The relation has the following additional functional dependencies:

FD1: Engineer_ID → Engineer_Name, Total_Repairs, Commission_Percentage, Engineer_phone

FD2: Comp_ID, Date_Assigned → Issue, Priority_Level, Service_Charge

FD3: Comp_ID → Customer_name, Customer_phone

FD4: Priority_Level → Service_Charge

FD5: Total_Repairs → Commission_Percentage

- a. Explain if this is in **1NF** or not. If not, apply normalization to decompose it to 1NF. [2]
- b. Explain if this is in **2NF** or not. If not, apply normalization to decompose it to 2NF. [4]
- c. Explain if this is in **3NF** or not. If not, apply normalization to decompose it to 3NF [4]

a. Already in 1NF as there are no composite, multivalued attributes, and no nested relations present.

b. NOT in 2NF due to partial dependency of FD2, FD3

Decomposing to 2NF:

Computer_Repair1 (Comp_ID, Date_Assigned, Issue, Priority_Level, Service_Charge)

Computer_Repair2 (Comp_ID, Customer_name, Customer_phone)

Computer_Repair(Engineer_ID, Engineer_Name, Total_Repairs, Commission_Percentage, Engineer_Phone)

c. NOT in 3NF due to TD of FD4, FD5

Computer_Repair3(Priority_Level, Service_Charge)

Computer_Repair4(Total_Repairs, Commission_Percentage)

Computer_Repair1 (Comp_ID, Date_Assigned, Issue, Priority_Level)

Computer_Repair2 (Comp_ID, Customer_name, Customer_phone)

Computer_Repair(Engineer_ID, Engineer_Name, Total_Repairs, Engineer_Phone)